MOHAMED ABUBAKAR

Computer Science Graduate | Software Developer | IT Professional

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PROFESSIONAL SUMMARY

Motivated Computer Science graduate with a strong interest in software development, full-stack engineering, database management, artificial intelligence, data science, cybersecurity, cloud computing, and computer networking. Experienced in developing academic and practical software projects, including rental websites, money market systems, tenant monitoring systems, study session platforms, and community disaster recovery systems. Passionate about solving real-world problems through technology, learning new tools, and developing secure, efficient, and user-friendly applications. Seeking an internship or entry-level opportunity to apply technical knowledge, gain professional experience, and contribute to organizational success.

EDUCATION

Bachelor's Degree in Computer Science
Machakos University
Graduation: 06/11/2026

Relevant areas of study:

- Programming and software engineering
- Data structures and algorithms
- Database management systems
- Computer networks and systems administration
- Artificial intelligence and machine learning
- Data science and analytics
- Cybersecurity and information systems

TECHNICAL SKILLS

Software Development

- Frontend and backend application development
- Web application design and development
- User authentication and access control
- Software testing and debugging
- API integration and application maintenance

Database and Data Engineering

- Database design and management

- Data processing and validation
- Data storage, retrieval, and integrity
- Data pipelines and reporting

Artificial Intelligence and Data Science

- Data analysis and visualization
- Machine learning fundamentals
- Data preprocessing and model evaluation

Networking and Cybersecurity

- Computer networking fundamentals
- Network troubleshooting and administration
- Application security and access management
- Data protection and security principles

Cloud Computing and DevOps

- Cloud application deployment fundamentals
- Software deployment and maintenance
- Backup, monitoring, and system reliability

Technologies and Tools
Proficient in programming languages such as Python, Java, JavaScript, PHP, and SQL; frameworks such as React, Node.js, and Django; databases such as MySQL and PostgreSQL; operating systems including Windows and Linux; cloud platforms such as AWS and Azure; development tools including Git, GitHub, and Visual Studio Code; and networking tools such as Cisco Packet Tracer and Wireshark.

PROJECT EXPERIENCE

1. Rental Website

Role: Full-Stack Developer

- Designed and developed a rental website to facilitate property listings and rental management.
- Implemented property search, filtering, and property information display.
- Worked on user authentication, property management, and database integration.
- Focused on usability, responsive design, and secure handling of user information.

2. Money Market System

Role: Software Developer

- Developed a money market system to support the management of financial and investment information.

- Worked on database design, financial data management, and transaction-related functionality.
- Applied security principles to user access and sensitive financial information.
- Focused on data accuracy, reliable processing, and efficient information retrieval.

3. Study Session Website

Role: Full-Stack Developer

- Developed a study session platform to support student scheduling, study group coordination, and learning activities.
- Worked on user management, session scheduling, and learning resource organization.
- Designed application features to improve collaboration and accessibility for students.

4. Community Disaster Recovery System

Role: Software Developer

- Developed a system concept and application functionality to support community disaster reporting and recovery coordination.
- Worked on disaster information management, emergency alerts, and relief resource tracking.
- Explored data analytics and the use of sensor-based technologies to support disaster monitoring and response planning.

5. Tenant Monitoring System

Role: Full-Stack Developer

- Developed a tenant management application for organizing tenant records, rent payments, lease information, and property occupancy.
- Worked on database management and the organization of property-related information.
- Focused on improving record accuracy and simplifying property management activities.

6. AI and Machine Learning System

Role: AI and Machine Learning Developer

- Worked on intelligent system development using data preprocessing and machine learning concepts.
- Explored model training, evaluation, pattern recognition, and predictive analysis.
- Applied data-driven approaches to automate tasks and support decision-making.

7. Hospital Management System

Role: Software Developer

- Worked on a hospital management application to support the organization of administrative and operational information.

- Focused on structured data management, accessibility, and efficient information retrieval.

8. Small-Scale Enterprise Management System

Role: Software Developer

- Developed an enterprise management application to support business record management and daily operational activities.
- Focused on organizing business information and improving workflow efficiency.

9. Procurement and Logistics System

Role: Software Developer

- Worked on a procurement and logistics application to support procurement records, inventory information, and resource tracking.
- Focused on data organization, process efficiency, and reliable information management.

10. Network Applications for Companies

Role: Network Application Developer

- Developed network-based application solutions to support organizational communication, information sharing, and collaboration.
- Focused on application connectivity, data exchange, access management, and reliable system operations.

PROFESSIONAL AND TECHNICAL INTERESTS

- Full-Stack and Software Engineering
- Data Engineering and Analytics
- Artificial Intelligence and Machine Learning
- Cybersecurity and Network Engineering
- Cloud Computing and DevOps
- Database Administration
- Robotics and Embedded Systems
- IT Support and Systems Administration

ACHIEVEMENTS & TRAINING

• Winner, Machakos University Innovation Day.

• Recognized by Kenyatta University for agricultural problem-solving initiatives.

• Participated in IEEE innovation challenges at Technical University of Kenya.

• Awarded by Zetech University for innovative embedded systems projects.

• Participated in agricultural innovation programs at Strathmore University.

• Web Development Certificate – Machakos Youth Centre.

• CompTIA Network+ Training.

• Cisco CCNA Training.

• Scrum Master Training.

• Project Management Skills Training.

• Training in digital data collection, survey administration, data quality control, respondent interviewing, and report preparation.

LEADERSHIP & COMMUNITY INVOLVEMENT

• Member, ZeroTech Company

• Member, MLSA Machakos University

• Member, GDG on Campus Machakos University

• Member, Cybersecurity Community

• Member, Computer Science Community

• Member, Cloud Native and Kubernetes Communities

WORK EXPERIENCE / TEACHING EXPERIENCE

Computer Science Lecturer
Machakos Youth Service | 04/09/2024-25/04/2025

- Taught or supported students in programming languages, software development, and computer science fundamentals.
- Guided practical programming exercises and software development projects.
- Supported students in developing logical reasoning, debugging, and problem-solving skills.

SOFT SKILLS

- Analytical and problem-solving skills
- Teamwork and collaboration
- Communication and presentation
- Time management and organization
- Adaptability and willingness to learn
- Attention to detail

PROFESSIONAL REFERENCES

Kuso Hussein Kuso 0723527119 – Environmental Specialist

Erick Omuya: 0727211203 – Lecturer at Machakos University

Henry Onyinkwa 0704547207 – SHA Specialist at Hola Referral Hospital Tana River

Mohamed Golo – 0742593773 – Father

Yatano Hussein Kuso – 0708208356 – Mather

Hadija Kuso – 0769208633 – Procurement and Logistics manager at KRA office in Mombasa